Exp. No: 9

Implementation of Magnetic-Field Boundary Conditions between Two Dielectric Media and Determination of Relative Permeability

Aim :

To implement magnetic-field boundary conditions between two media and determine the relative permeability of the unknown medium.

Objectives:

1. To resolve magnetic-field intensity into normal and tangential components at a material interface.

2. To apply the tangential H-field and normal B-flux-density boundary conditions.

3. To determine the relative permeability of an unknown medium from specified boundary data.

4. To verify the magnetic boundary conditions numerically using MATLAB.

Test Case (i) – Magnetic Boundary Conditions and Determination of the Relative Permeability

MATLAB Program:

```
clc;

clear;

close all;

%% Magnetic-Field Boundary Conditions between Two Media mu0 = 4pi1e-7;

% Input data %

H/m mur1 = input('Enter relative permeability of medium 1: ');

H1t = input('Enter tangential H-field H1t (A/m): ');
```

```
B1n = input('Enter normal flux density B1n (T): ');
 H2n_meas = input('Enter measured/assigned normal H-field H2n (A/m): ');
mur2_ref = input('Enter reference relative permeability of medium 2: ');
% Permeability of the two media mu1 = mu0*mur1; mu2_ref = mu0*mur2_ref;
% Boundary conditions for zero surface current H2t = H1t; B2n = B1n;
 % Determine relative permeability from measured
 H2n mur2_calc = B2n/(mu0*H2n_meas);
 % Theoretical normal H-field using the reference permeability
 H2n_theory = B2n/mu2_ref;
% Percentage deviation dev = abs(mur2_calc-mur2_ref)/abs(mur2_ref)*100;
 fprintf('\n--- Magnetic Boundary Condition Results ---\n');
fprintf('H2t (theoretical) = %.6f A/m\n',H2t);
 fprintf('B2n (theoretical) = %.6e T\n',B2n);
 fprintf('H2n (theoretical) = %.6f A/m\n',H2n_theory);
 fprintf('Calculated mur2
  %.6f\n',mur2_calc); fprintf('Reference mur2
 =%.6f\n',mur2_ref);
 fprintf('Percentage deviation = %.4f %%\n',dev);
% Parameter sweep: effect of relative permeability on normal
 H-field mur = linspace(1,10,100);
Hn = B1n./(mu0*mur); figure;
 plot(mur,Hn,'LineWidth',2); grid on;
xlabel('Relative permeability, \mu_r'); ylabel('Normal magnetic field H_n (A/m)');
 title('Normal H-field versus Relative Permeability');
```

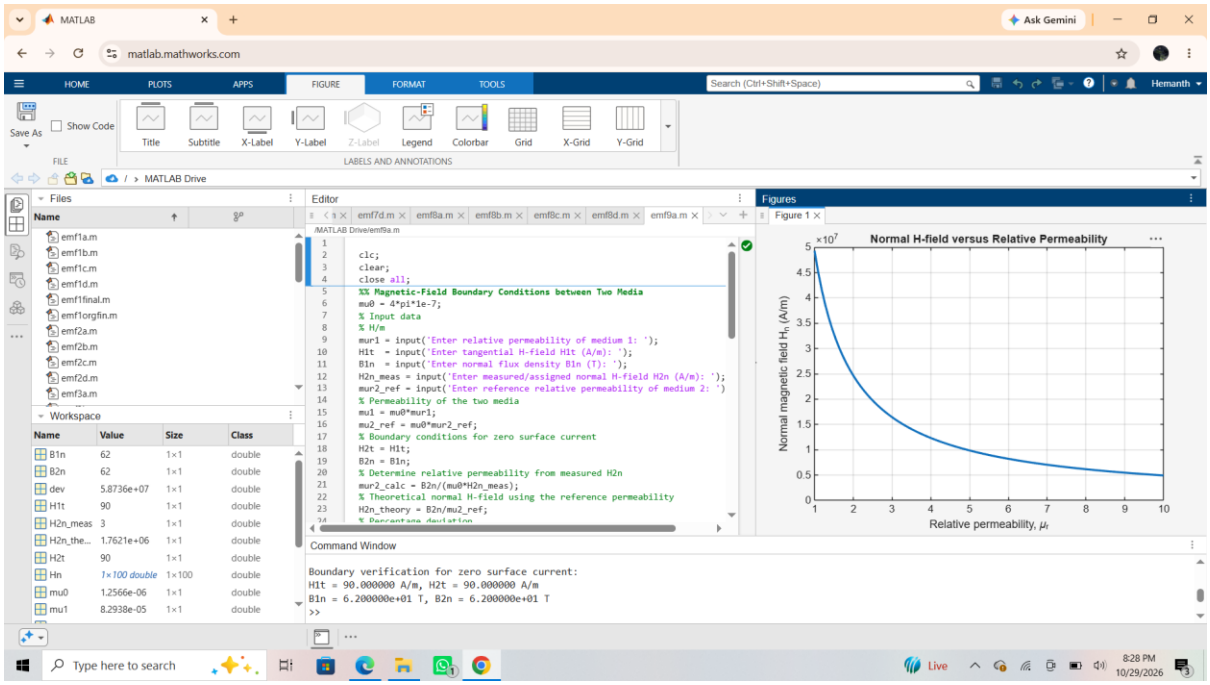
% Verification of tangential H continuity and normal B continuity

fprintf('\nBoundary verification for zero surface current:\n');

fprintf('H1t = %.6f A/m, H2t = %.6f A/m\n',H1t,H2t);

fprintf('B1n = %.6e T, B2n = %.6e T\n',B1n,B2n);

Output:




Test Case (ii) – Field Components and Refraction Angle at the Interface of Two Magnetic Media

MATLAB Program:

clc; clear;

mu0 = 4*pi*1e-7;

H1 = input('Enter magnitude of H1 in medium 1 (A/m): ');

th1 = input('Enter angle of H1 with the normal (deg): ');

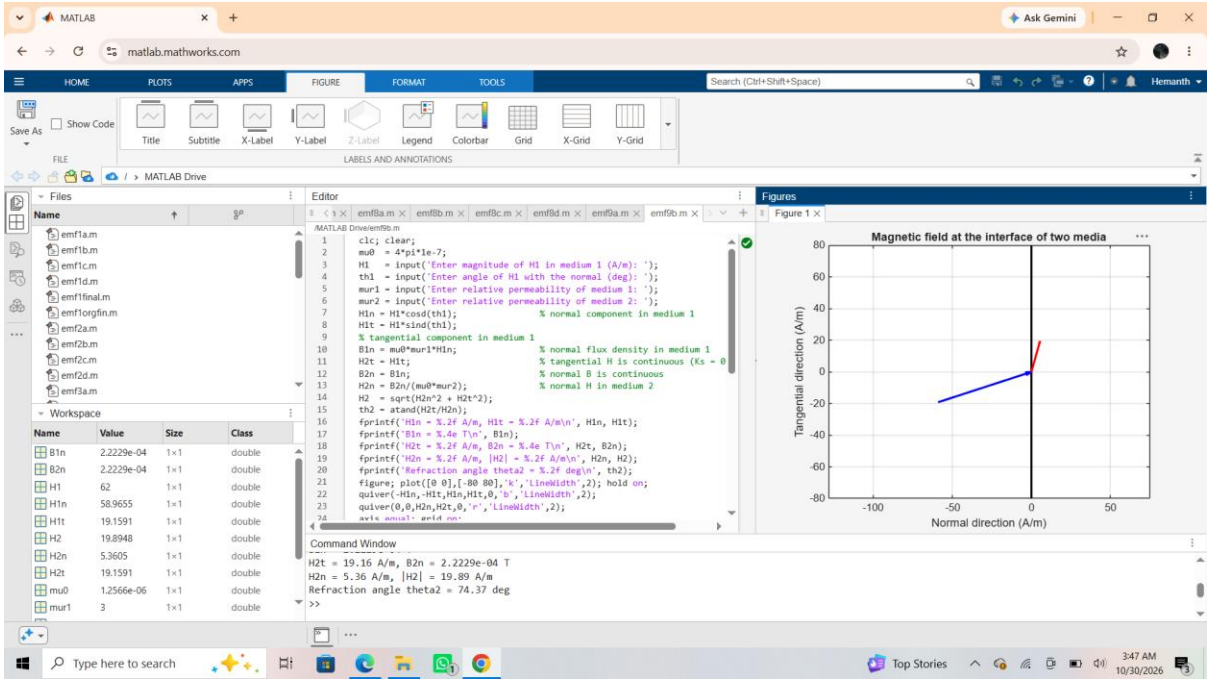
mur1 = input('Enter relative permeability of medium 1: ');

mur2 = input('Enter relative permeability of medium 2: ');

H1n = H1*cosd(th1); % normal component in medium 1 H1t = H1*sind(th1);

```
% tangential component in medium 1 B1n = mu0*mur1*H1n;
% normal flux density in medium 1 H2t = H1t;
% tangential H is continuous (Ks = 0) B2n = B1n;
% normal B is continuous H2n = B2n/(mu0*mur2);
% normal H in medium 2 H2 = sqrt(H2n^2 + H2t^2);
th2 = atand(H2t/H2n);
fprintf('H1n = %.2f A/m, H1t = %.2f A/m\n', H1n, H1t);
fprintf('B1n = %.4e T\n', B1n);
fprintf('H2t = %.2f A/m, B2n = %.4e T\n', H2t, B2n);
fprintf('H2n = %.2f A/m, |H2| = %.2f A/m\n', H2n, H2);
fprintf('Refraction angle theta2 = %.2f deg\n', th2);
figure; plot([0 0],[-80 80],'k','LineWidth',2); hold on;
quiver(-H1n,-H1t,H1n,H1t,0,'b','LineWidth',2);
quiver(0,0,H2n,H2t,0,'r','LineWidth',2); axis equal; grid on;
xlabel('Normal direction (A/m)'); ylabel('Tangential direction (A/m)');
title('Magnetic field at the interface of two media');
```

Output:



Test Case (iii) – Variation of the Refraction Angle with the Angle of Incidence for Different Permeabilities

MATLAB Program:

```
clc; clear;

th1 = linspace(0,89,179);

mur1 = 1;

mur2 = [2 5 10];

figure; hold on; for k = 1:3 th2 = atand((mur2(k)/mur1)*tand(th1));

 % tan(th2) = (mur2/mur1)*tan(th1) plot(th1,th2,'LineWidth',2);

end plot(th1,th1,'k--');

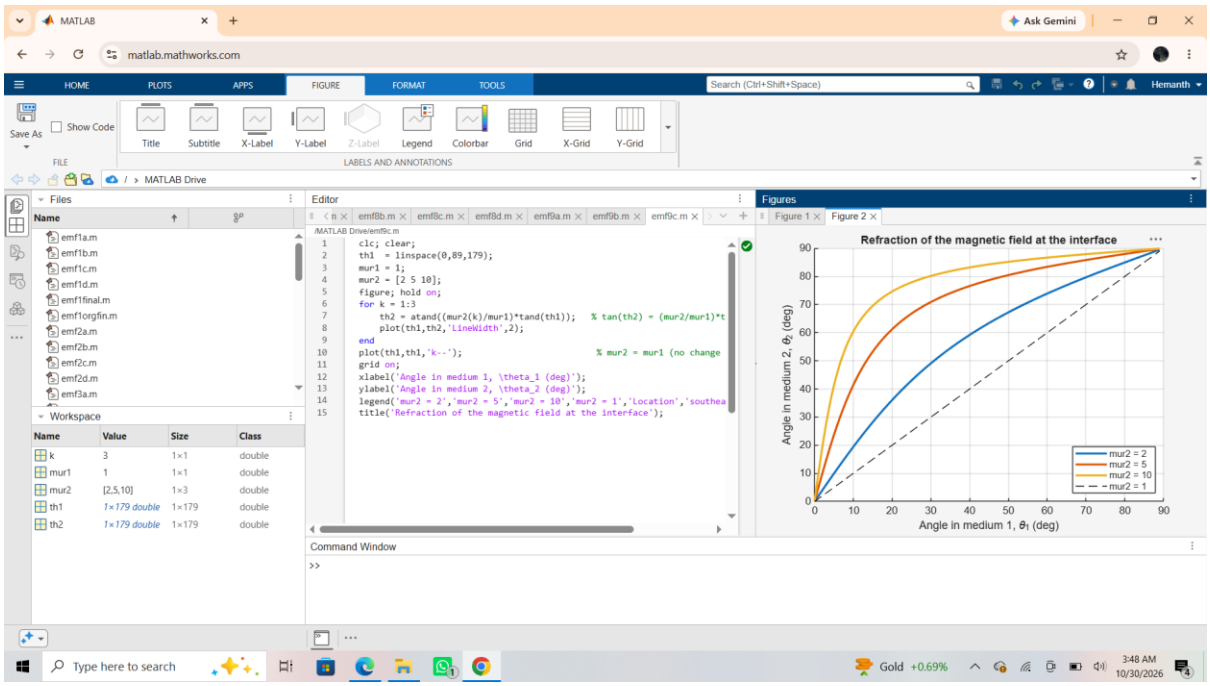
 % mur2 = mur1 (no change of direction) grid on;

xlabel('Angle in medium 1, \theta_1 (deg)'); ylabel('Angle in medium 2, \theta_2 (deg)');

 legend('mur2 = 2','mur2 = 5','mur2 = 10','mur2 = 1','Location','southeast');

 title('Refraction of the magnetic field at the interface');
```

Output:



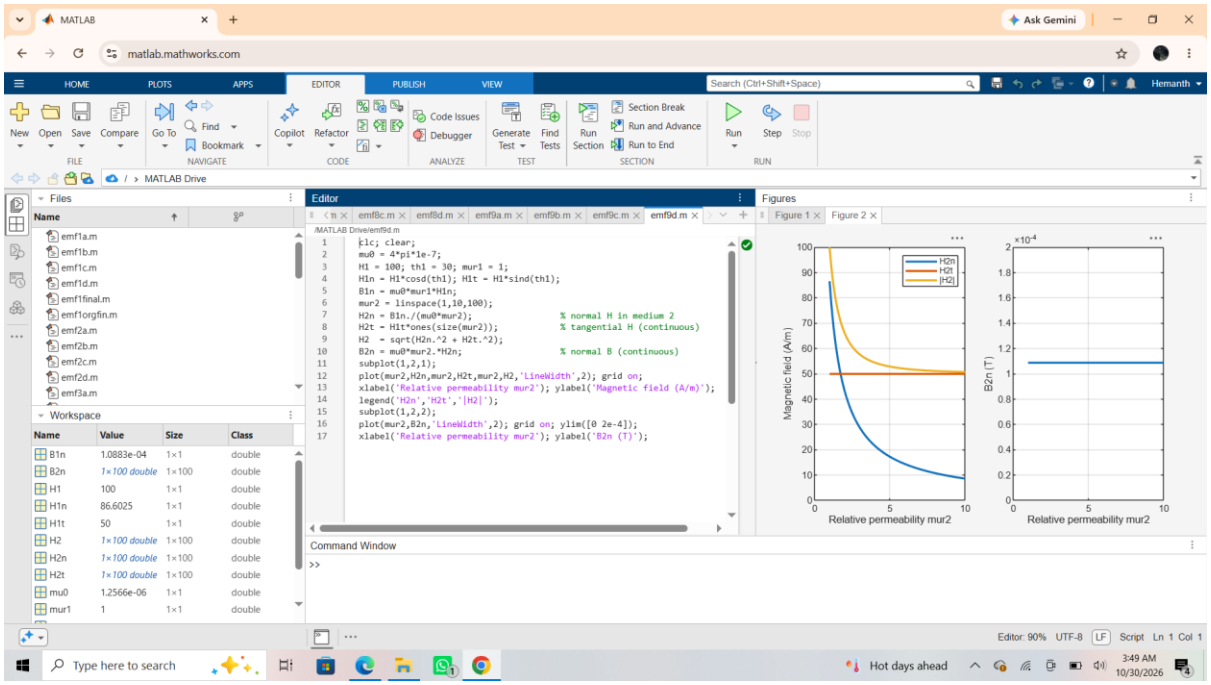
Test Case (iv) – Effect of the Relative Permeability on the Field Components in Medium 2

MATLAB Program:

```
clc; clear;
 mu0 = 4*pi*1e-7;
 H1 = 100; th1 = 30;
 mur1 = 1;
 H1n = H1*cosd(th1);
 H1t = H1*sind(th1);
B1n = mu0*mur1*H1n;
 mur2 = linspace(1,10,100);
 H2n = B1n./(mu0*mur2);          % normal H in medium 2
 H2t = H1t*ones(size(mur2));       % tangential H (continuous)
 H2 = sqrt(H2n.^2 + H2t.^2);
 B2n = mu0*mur2.*H2n;                % normal B (continuous)

 subplot(1,2,1);
plot(mur2,H2n,mur2,H2t,mur2,H2,'LineWidth',2); grid on;
xlabel('Relative permeability mur2'); ylabel('Magnetic field (A/m)');
 legend('H2n','H2t','|H2|'); subplot(1,2,2);
plot(mur2,B2n,'LineWidth',2); grid on; ylim([0 2e-4]);
 xlabel('Relative permeability mur2'); ylabel('B2n (T)');
```

Output:




| S. No. | Objective | Assessment Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weighting (%) | Marks given | Marks obtained |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Identify magnetic media and boundary orientation | Identify magnetic media and boundary orientation | Correctly identifies permeability, normal/tangential components and surface current assumptions. | Setup is correct with minor omissions. | Needs help identifying components or assumptions. | Uses incorrect interface definitions. | 25 | 2.5 | |
| 2 | Apply magnetic boundary relations | Apply magnetic boundary relations | Correctly applies continuity of normal B and | Relations are applied with | . Partly correct application requires guidance. | Boundary relations are incorrect. | 25 | 2.5 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | the tangential H relation. | minor errors. | | | | | |
| 3 | Determine permeability and validate fields | Determine permeability and validate fields | Calculates required quantities and verifies the boundary relations consistently | Results are mostly correct with minor discrepancies. | Validation is incomplete. | Results are invalid or unverified. | 30 | 3 | |
| 4 | Explain and document findings | Explain and document findings | Explains magnetic-field behaviour and records assumptions, units and conclusion. | Documentation is clear with minor omissions. | Interpretation or documentation is incomplete. | Cannot explain or report results. | 20 | 2 | |
| Total | | | | | | | 100 | 10 | |